

Stanford University
Political Science 350D
Problem Set 2
Due April 13, 2015

1. Suppose $y \sim \text{Normal}(0, \sigma^2)$.

(a) What is the information $\mathcal{J}(\sigma^2)$ in y ?

Solution: The log likelihood and its first two derivatives are

$$\begin{aligned}\log p(y|\sigma^2) &= -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \sigma^2 - \frac{y^2}{2\sigma^2} \\ \frac{\partial \log p(y|\sigma^2)}{\partial \sigma^2} &= -\frac{1}{2\sigma^2} + \frac{y^2}{2\sigma^4} \\ \frac{\partial^2 \log p(y|\sigma^2)}{\partial (\sigma^2)^2} &= \frac{1}{2\sigma^4} - \frac{y^2}{\sigma^6}\end{aligned}$$

Taking expectations, $\mathbb{E}(y^2|\sigma^2) = \sigma^2$, so

$$\mathcal{J}(\sigma^2) = -\mathbb{E}\left(\frac{\partial^2 \log p(y|\sigma^2)}{\partial (\sigma^2)^2}\right) = -\frac{1}{2\sigma^4} + \frac{\sigma^2}{\sigma^6} = \frac{1}{2\sigma^4}.$$

(b) What is Jeffreys' prior for σ^2 ?

Solution: Jeffreys' prior is

$$p(\sigma^2) \propto \sqrt{\mathcal{J}(\sigma^2)} = \sqrt{\frac{1}{2\sigma^4}} \propto \frac{1}{\sigma^2}.$$

This is an improper prior, since $\int_0^\infty dx/x = \infty$. It is in the form of an Inverse Gamma(0, 0) or a Scaled Inverse Chi-squared(0, 0) distribution.

(c) Suppose you observe $y = (-2, 3, 0, -1, 2)$. What is the posterior distribution of σ^2 using the prior from part (b)?

Solution: With $p(\sigma^2) \propto 1/\sigma^2$ and

$$p(y_1, \dots, y_n | \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left\{-\frac{\sum_{i=1}^n y_i^2}{2\sigma^2}\right\},$$

the posterior distribution is

$$p(\sigma^2 | y_1, \dots, y_n) \propto p(\sigma^2) p(y_1, \dots, y_n | \sigma^2) \propto (\sigma^2)^{-n/2-1} e^{-S(0)/2\sigma^2},$$

where $S(\mu) = \sum_{i=1}^n (y_i - \mu)^2$. This is an Inverse Gamma($n/2, S(0)/2$) or a Scaled Inverse Chi-squared($n, S(0)/n$) distribution. In this case $n = 5$ and $S(0) = 18$, so the posterior is either Inverse Gamma(2.5, 9) or Scaled Inverse Chi-square(5, 3.6).

Stanford University
Political Science 350D
Problem Set 2 (Continued)

- (d) Compute the posterior mean and mode of σ^2 using your answer from part (c).

Solution: The posterior mean and variance are given in Table A.1 of *BDA3*. The posterior mean is $18/3 = 6$ and the posterior mode is $18/7 \doteq 2.57$.

2. Exercise 2.11.8 in *BDA3*. A random sample of n students is drawn from a large population, and their weights are measured. The average weight of the n sampled students is $\bar{y} = 150$ pounds. Assume the weights in the population are normally distributed with unknown mean θ and known standard deviation 20 pounds. Suppose your prior distribution for θ is normal with mean 180 and standard deviation 40.

- (a) Give your posterior distribution for θ .

Solution: Using

$$\begin{aligned}\mathbb{E}(\theta|y) &= \frac{(1/40^2)180 + (n/20^2)150}{(1/40^2) + (n/20^2)} = \frac{180 + 600n}{1 + 4n} \\ \frac{1}{\mathbb{V}(\theta|y)} &= \frac{1}{40^2} + \frac{n}{20^2} = \frac{1600}{1 + 4n},\end{aligned}$$

we obtain

$$\theta|y \sim \text{Normal}\left(\frac{180 + 600n}{1 + 4n}, \frac{1600}{1 + 4n}\right).$$

- (b) A new student is sampled at random from the same population and has a weight of $\tilde{y}$ pounds. Give a posterior predictive distribution for $\tilde{y}$.

Solution: The posterior predictive distribution for $\tilde{y}$ is normal with mean $\mathbb{E}(\theta|\bar{y})$ and variance $\mathbb{V}(\theta|\bar{y}) + 20^2$. Using the answer to part (a),

$$\tilde{y}|\bar{y} \sim \text{Normal}\left(\frac{180 + 600n}{1 + 4n}, \frac{1600}{1 + 4n} + 20^2\right).$$

- (c) For $n = 10$, give a 95% posterior interval for θ and a 95% posterior predictive interval for $\tilde{y}$.
(d) Do the same for $n = 100$.

Solution:

Stanford University
Political Science 350D
Problem Set 2 (Continued)

```
> ybar <- 150; sigma <- 20; mu.0 <- 180; tau.0 <- 40
> solution <- function(n) list(theta=qnorm(c(.025,.975),
+   mean=(180+600*n)/(1+4*n), sd=sqrt(1600/(1+4*n))),
+   y.tilde=qnorm(c(.025,.975), mean=(180+600*n)/(1+4*n),
+   sd=sqrt(1600/(1+4*n) + 20^2)))
> solution(n=10)                # part (c)
$theta
[1] 138.4879 162.9755

$y.tilde
[1] 109.6648 191.7987

> solution(n=100)              # part (d)
$theta
[1] 146.1598 153.9899

$y.tilde
[1] 110.6805 189.4691
```

3. Exercise 2.11.13 in *BDA3*.

Table 2.2 gives the number of fatal accidents and deaths on scheduled airline flights per year over a ten-year period.

- (a) Assume that the numbers of fatal accidents in each year are independent with a Poisson(θ) distribution. Set a prior distribution for θ and determine the posterior distribution based on the data from 1976 through 1985. Under this model, give a 95% predictive interval for the number of fatal accidents in 1986.

Solution: If $y_1, \dots, y_n$ are i.i.d. Poisson(θ),

$$p(y_i|\theta) = \frac{\theta^{y_i} e^{-\theta}}{y_i!} \quad (i = 1, \dots, n)$$

You may choose whatever prior you wish, but Jeffreys' prior is $p(y) \propto \theta^{-1/2}$ for $\theta > 0$. If you use Jeffreys' prior, the posterior distribution is

$$p(\theta|y) \propto \theta^{-1/2} \theta^{n\bar{y}} e^{-n\theta} = \theta^{n\bar{y}-1/2} e^{-n\theta}$$

which is the kernel of a Gamma($n\bar{y} + 1/2, n$) distribution. You can obtain a posterior predictive interval by simulating M draws $\theta \sim \text{Gamma}(n\bar{y} + 1/2, n)$ random variables

Stanford University
Political Science 350D
Problem Set 2 (Continued)

and then simulating $\tilde{y} \sim \text{Poisson}(\theta)$ from each and computing the 2.5% and 97.5% quantiles.

```
> df <- data.frame(year=1976:1985,
+   accidents=c(24,25,31,31,22,21,26,20,16,22),
+   deaths=c(734,516,754,877,814,362,764,809,223,1066),
+   rate=c(0.19,0.12,0.15,0.16,0.14,0.06,0.13,0.13,0.03,0.15))
> # part (a)
> ybar <- mean(df$accidents); n <- nrow(df); M <- 10000
> theta <- rgamma(M, shape=n*ybar-1/2, rate=n)
> ytilde <- rpois(M, lambda=theta)
> quantile(ytilde, probs=c(.025, .975))
2.5% 97.5%
14 34
```

- (b) Assume that the numbers of fatal accidents in each year follow independent Poisson distributions with a constant rate and an exposure in each year proportional to the number of passenger miles flow. Set a prior distribution for θ and determine the posterior distribution based on the data for 1976–1985. (Estimate the number of passenger miles flown in each year by dividing the appropriate columns of Table 2.2 and ignoring round-off errors.) Give a 95% predictive interval for the number of fatal accidents in 1986 under the assumption that 8×10^{11} passenger miles are flow that year.

Solution: The model is now that $y_1, \dots, y_n$ are i.n.i.d. $\text{Poisson}(x_i\theta)$ where x_i = miles flow in year i . Using the same prior as in part (a) ($\theta \propto \theta^{-1/2}$), the posterior is now

$$p(\theta|y) \propto \theta^{-1/2} \prod_{i=1}^n (x_i\theta)^{y_i} e^{-x_i\theta} \\ \propto \theta^{n\bar{y}-1/2} e^{-n\bar{x}\theta}$$

which is the kernel of a $\text{Gamma}(n\bar{y}+1/2, n\bar{x})$ distribution. Here is R code to simulate M draws from the predictive distribution for $\tilde{x} = 8 \times 10^{11}$. (Remember that miles flown is measured in hundreds of millions of miles!)

```
> df$miles <- with(df, deaths / rate) # miles flown / 10^8
> xbar <- mean(df$miles); xtilde <- 8e3
> theta <- rgamma(M, shape=n*ybar+1/2, rate=n*xbar)
> ytilde <- rpois(M, lambda=xtilde*theta)
> quantile(ytilde, probs=c(.025, .975))
```

Stanford University
Political Science 350D
Problem Set 2 (Continued)

2.5%	97.5%
22	46

- (c) Repeat (a) above, replacing ‘fatal accidents’ with ‘passenger deaths.’

Solution:

```
> ybar <- mean(df$deaths)
> theta <- rgamma(M, shape=n*ybar-1/2, rate=n)
> ytilde <- rpois(M, lambda=theta)
> quantile(ytilde, probs=c(.025, .975))
2.5% 97.5%
637 746
```

- (d) Repeat (b) above, replacing ‘fatal accidents’ with ‘passenger deaths.’

Solution:

```
> theta <- rgamma(M, shape=n*ybar+1/2, rate=n*xbar)
> ytilde <- rpois(M, lambda=xtilde*theta)
> quantile(ytilde, probs=c(.025, .975))
2.5% 97.5%
905 1035
```

- (e) In which of the cases (a)–(d) above does the Poisson model seem more or less reasonable? Why? Discuss based on general principles, without specific reference to the numbers in Table 2.2.

Solution: The Poisson assumption (independent events occurring at constant rates in time) makes more sense for accidents than deaths, since a fatal airline accident usually kills most or all of the passengers (“clumping”). Miles flown probably isn’t the best exposure variable for accidents (since it is passenger miles, rather than aircraft miles—accidents are likely to be proportional to the latter) and the use of the exposure variable greatly exacerbates the lack of independence in deaths.